

# Grokking in LLM Pretraining? Monitor Memorization-to-Generalization without Test

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## Abstract

This paper presents *the first study of grokking in practical LLM pretraining*. Specifically, we investigate when an LLM memorizes the training data, when its generalization on downstream tasks starts to improve, and what happens if there is a lag between the two. Unlike existing works studying when a small model generalizes to limited and specified tasks during thousands epochs’ training on algorithmic data, we focus on a practical setting for LLMs, i.e., near single-pass pretraining of next-token prediction on a cross-domain, large-scale corpus, and generalization on diverse benchmark tasks covering math/commonsense reasoning, code generation, and domain-specific retrieval. Our study, *for the first time, verifies that grokking still emerges in pretraining mixture-of-experts (MoE) LLMs*, though different local data groups may enter their grokking stages asynchronously due to the heterogeneity of their distributions and attributions to others. To find a mechanistic interpretation of this local grokking, we investigate the dynamics of training data’s pathways (i.e., expert choices across layers in MoE). Our primary discovery is that *the pathways evolve from random, non-smooth across layers, instance-specific to more structured and transferable across samples*, despite the converged pretraining loss. This depicts a transition from memorization to generalization. Two novel metrics are developed to quantify these patterns: one computes the pathway similarity between samples, while the other measures the consistency of aggregated experts between subsequent layers for each sample. These training data based metrics induce near zero cost but can faithfully track and monitor the generalization of LLMs on downstream tasks, reducing reliance on costly instruction tuning and benchmark evaluations. We also ground our findings in a theoretical analysis of one-layer MoE, showing that more structured pathways improve the generalization bound.

## 1 Introduction

Large language models (LLMs) with Transformer [30] architectures have demonstrated that pretraining for a general-purpose task such as autoregressive prediction on a large corpus can gain powerful generalization across diverse downstream tasks and domains [3, 29, 2, 36]. Capabilities such as prompting [32], reasoning [10, 1], and in-context learning also emerge during training. However, a counter-intuitive “grokking” [24, 22, 28, 18] has been widely observed in training Transformer models: the generalization starts to improve or even sharply rises long after training loss converged or plateaued—an indicator of overfitting in conventional machine learning. While grokking indicates a transition from memorization to generalization [11], it challenges our fundamental understanding of training dynamics and the underlying mechanism of generalization.

Despite recent attempts to explain the generalization induced by grokking, they have been largely confined to small models trained for hundreds to thousands of epochs on algorithm-synthesized small-scale data with generalization to limited and highly specified tasks [24, 15, 22, 19, 12, 5, 25]. Instead, LLMs’ pre-training often takes only one epoch on a web-scale corpus of heterogeneous,

cross-domain data that usually contains noise, imbalance, and redundancy [3, 36]. Hence, LLMs’ learning dynamics can vary drastically across different data, leaving whether grokking still emerges in LLM pretraining an open problem. It is also unclear whether and when different data are memorized with converged loss, and, whether they are memorized at the same time. Moreover, without the repeated replay of the same data and a decrease in loss, the conversion from memorizing training tokens to generalizing on downstream tasks remains mysterious: When does the model start to generalize to different tasks? Is memorization a prerequisite? How does generalization performance change after memorization?

Unfortunately, these questions cannot be answered by the dynamics of pretraining loss since the generalization performance can still vary after the loss plateaued, as observed in previous studies of grokking [24, 22]. For LLMs, it is also practically expensive to monitor its generalization performance during pretraining, since the model has to be finetuned to gain instruction-following capability before being evaluated on downstream tasks. Hence, developing an efficient-to-compute metric to accurately track the generalization can offer critical insights on better understanding and improving the pretraining of LLMs.

**Main Contributions** In this paper, we investigate the training dynamics and grokking in LLM pretraining for a mechanistic interpretation of its emergent generalization on downstream tasks. Our study is conducted on the open-sourced pretraining checkpoints of a 7B mixture-of-experts (MoE) LLM, OLMoE [21]. While our study mainly focuses on OLMoE, its pretraining and evaluation setups are common among other LLMs. For the first time, we verify the existence of local grokking during LLM pretraining. However, unlike the previously observed global and synchronous grokking for most data [24, 19, 22], *LLM memorizes domains/groups of training data at different training steps and takes varying amounts of steps to generalize to related benchmark tasks*. The starting time and lasting steps of such local grokking vary across data groups. As a result, generalization performance is usually unstable during the earlier pretraining stages but starts to steadily improve once sufficient data have been memorized. This local difference also reflects data difficulty: the later memorized data group often takes longer to generalize.

Another primary challenge is to explain how the continual pretraining after memorization completed leads to the delayed sharp rise of the generalization performance. To investigate the mechanism of memorization-to-generalization transition, we explore internal states of LLMs track their major changes after loss converges. Specifically, we study how the pathways (the expert choices across layers for training samples) in OLMoE change. We introduce two metrics to measure their complexity how it changes over time: (1) the edit distance between different samples’ pathways; and (2) the consistency of selected experts’ embedding between consecutive layers for each sample.

Our empirical analysis on four domains reveals a prominent trend of the pathway complexity during grokking: the edit distance declines and the consistency increases, despite the converged training loss and more data being learned. This change in pathway complexity reflects a transition to smarter memorization: it keeps discovering more transferable knowledge across samples to encode data more efficiently. *It explains why generalization improves with plateaued training loss: the model keeps finding more generalizable and shareable structures to memorize training data*. Additionally, we provide a theoretical explanation relating the pathway complexity to a generalization bound of a one-layer MoE. Practically, the two metrics show strong correlations with the evaluation results on standard benchmarks after instruction finetuning. Therefore, they provide a zero-cost, finetuning and benchmark evaluation-free tool to monitor generalization during pretraining, which is critical to LLM developers and practitioners. These discoveries take the first step toward bridging the gaps of understanding grokking and memorization-to-generalization transition in LLM pretraining. Our key findings and contributions can be summarized as:

- Grokking still occurs during the near single-pass pretraining of practical-scale LLMs, but it is local and asynchronous for different data groups, unlike global grokking for all data in previous works.
- Grokking’s memorization-to-generalization transition can be explained by the dynamics of pathways in MoE: pathway similarity between samples and consistency across layers increase. A theoretical connection is also built between the pathway complexity and a generalization bound.
- The two metrics we developed to measure pathway complexity provide a finetuning and evaluation-free tool with almost zero cost to monitor the generalization during LLM pretraining.

## 2 Related Work

Previous studies on grokking have primarily focused on relatively small models and simplistic tasks. Grokking was first identified by Power et al. [24], who observed the phenomenon using a two-layer transformer trained on a small-scale algorithmic dataset. Subsequent investigations have extended these findings to shallow transformers and densely connected networks [22, 23, 19, 16, 6], and CNNs [12]. Notably, all aforementioned studies employ multi-epoch training paradigms, which differ substantially from the one-pass training approach typical in LLM pretraining. Although Lv et al. [18] examined grokking behavior using a 162M parameter transformer pretrained on 40 billion tokens for a copying task, the scale of their model, dataset, and the simplicity of the task remain distant from practical, real-world scenarios. In contrast, our study is the first to investigate grokking in a 7-billion parameter LLM across diverse and practical tasks, including mathematical reasoning, code generation, commonsense understanding, and domain-specific knowledge retrieval.

Several works have focused on revealing the mechanisms behind grokking behavior. Liu et al. [15] demonstrated in a simplified setting that grokking results from structured representations emerging during training. Nanda et al. [22] further revealed that grokking can be attributed to the gradual amplification of structured mechanisms encoded within model weights. Wang et al. [31] showed that even small Transformers trained from scratch on synthetic reasoning tasks exhibit a grokking-driven emergence of implicit reasoning, where generalization arises only after extended overfitting and internal structure formation. Using a single-layer ReLU network, Merrill et al. [19] associated grokking with subnetwork sparsity. Meanwhile, Humayun et al. [12] observed that decision boundaries become smoother during the grokking phase in CNN classification tasks. Beyond interpretability-focused research, Notsawo Jr et al. [23] identified oscillatory patterns in early training loss as predictors of subsequent grokking phenomena. Distinct from these works, our research provides an explanation of grokking within significantly larger LLMs through an interpretative analysis of MoE architectures. Additionally, we introduce practical indicators for monitoring generative behavior throughout LLM pretraining.

## 3 Grokking (Delayed Generalization) in LLM Pretraining

While most existing works study grokking of small models on specified algorithmic tasks, investigating similar phenomena in LLM pretraining poses several new unique challenges, which inherently motivate several designs in our experimental settings.

- **Training-test evaluation gap:** most LLMs are trained for the next token(s) prediction while evaluated on diverse benchmark tasks, making the conventional comparison between training and test loss/accuracy invalid. Hence, we need to adapt the definition of grokking to the LLM setting.
- **Heterogeneous, web-sourced, cross-domain training data:** While previous grokking studies focus on clean training data with the same distribution/format, most LLMs are pretrained on heterogeneous, cross-domain, web-sourced data that contain redundant, contaminated, or imbalanced samples. Hence, their loss may converge at different rates and pose distinct effects on downstream tasks. To address these challenges, we carefully filter and group the data in our analysis.
- **Generalization to diverse downstream tasks:** Previous studies of grokking focus on generalization to highly-specified and limited tasks, while practical LLMs are expected to generalize to arbitrary tasks or domains via prompting. Hence, LLMs’ generalization behaviors may vary drastically across different downstream tasks/benchmarks and difficulty levels during pretraining.
- **Near single-pass pretraining:** Most LLMs’ pretraining only takes approximately one epoch, so the model has been trained only once on each sample, and its training loss may even converge before visiting later data. However, such “memorization” of unseen data is not caused by overfitting or repeated replay. Instead, it reflects the interpolation or composition of learned data, which might be counted as “generalization” in previous works but differs from the emergent generalization of LLMs on new tasks.

### 3.1 Study Training Dynamics of LLM Pretraining

**Memorization is measured by the pretraining objective.** Next token prediction (NTP) is the most widely adopted pretraining objective for LLMs. Let the pretraining corpus be  $D_{\text{train}}$ , it aims to